

PRST (18/11/2021).

Ex ① $f_{X,Y} = \begin{cases} cx+1 & x,y > 0 \\ & x+y < 1 \\ 0 & \text{o.w.} \end{cases}$

① $c = ?$

② $f_X, f_Y ?$

③ $P(Y < 2X^2)$.

$$\int_0^1 \int_0^{1-x} f_{X,Y}(x,y) dy dx = 1$$

$$\Rightarrow \int_0^1 \int_0^{1-x} (cx+1) dy dx = 1$$

$$\Rightarrow \int_0^1 (cx+1)(1-x) dx = 1$$

$$\Rightarrow \frac{1}{2} + \frac{1}{6}c = 1$$

$$\Rightarrow c = 3.$$

$$f_X = \int_0^{1-x} (3x+1) dy = (3x+1)(1-x) \quad 0 \leq x \leq 1$$

$$f_X(x) = \begin{cases} (3x+1)(1-x) & 0 \leq x \leq 1 \\ 0 & \text{o.w.} \end{cases}$$

$$f_Y = \int_0^{1-y} (3x+1) dx = \frac{1}{2}(1-y)(5-3y) \quad 0 \leq y \leq 1$$

$$f_Y(y) = \begin{cases} \frac{1}{2}(1-y)(5-3y) & 0 \leq y \leq 1 \\ 0 & \text{o.w.} \end{cases}$$

$$P(Y < 2X^2) = \int_0^1 \int_0^{\min(2x^2, 1-x)} (3x+1) dy dx$$

$$= \int_0^{\frac{1}{2}} (3x+1) \cdot \min(2x^2, 1-x) dx + \int_{\frac{1}{2}}^1 (3x+1)(1-x) dx$$

$$= \int_0^{\frac{1}{2}} (3x+1) \cdot 2x^2 dx + \frac{1}{2}$$

$$= \frac{53}{96}$$

Ex 2 $f_{X,Y} = \frac{1}{6} e^{-(2x+3y)} \quad x,y > 0$

① X and Y ind or not

② $E(Y | X > 2)$

③ $P(X > Y)$

Quiz-4 sol

$$X_1, X_2, X_3$$

$$f(x) = \frac{8}{x^3}, x \geq 2$$

$$P(X_1 > x) = 1 - F(x)$$

$$P(X_1 > 4) = P(X_2 > 4, X_3 > 4)$$

$$= P(X_1 > 4) \cdot P(X_2 > 4) \cdot P(X_3 > 4)$$

$$= (1 - F(x))^n$$

$$(3) P(x < 1, y < 2)$$

$$f_{xy} = \frac{2e^{-2x}}{3} \times \frac{3e^{-3y}}{3}, x > 0, y > 0$$

$$E(y) = \frac{1}{3}$$

$$P(x > y) = \int_0^\infty \int_y^\infty \frac{2e^{-2x}}{3} \times \frac{3e^{-3y}}{3} dx dy = \frac{3}{5}$$

$$= P(x_1 > 4) P(x_2 > 4) P(x_3 > 4) = P(x_1 > 4)^3 = (1 - F(4))^3$$

$$F(x) = \int_2^x \frac{8}{t^3} dt$$

$$= 1 - \frac{4}{x^2}$$

$$(1 - F(4))^3 = \left(\frac{4}{4^2}\right)^3 = \left(\frac{1}{4}\right)^3 = \frac{1}{64}$$

$$P(X(1) < x)$$

$$P(X(n) < x) = P(X_i < x)^n = F(x)^n$$

$$X \sim \log N(\mu, \sigma)$$

$$\mu = 5.149$$

$$\sigma = 0.737$$

$$\log c = 2$$

$$P(X < c) = 0.05$$

$$P\left(\frac{\log X}{\sigma} < z\right) = 0.05$$

$$P\left(\frac{z - \mu}{\sigma} < \frac{z - \mu}{\sigma}\right) = 0.05$$

$$\Rightarrow \Phi\left(\frac{z - \mu}{\sigma}\right) = 0.05$$

$$\frac{z - \mu}{\sigma} = -1.645$$

$$\Rightarrow z = 3.937$$

$$\log c = 3.937$$

$$c = e^{3.937}$$

Ex 3.

$$f_X(x) = \begin{cases} 2x & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

$$0 \leq x \leq 1$$

$$0 \leq y \leq 1$$

$$Y \sim U(-x, x)$$

given $X = x$,

$$(1) f_{xy}$$

$$(2) f_y$$

$$(3) P(|Y| < x^3)$$

$$-x \leq y \leq x$$

$$f_{Y|X} = \begin{cases} \frac{1}{2x} \\ 0 \end{cases}$$

$$-x \leq Y \leq x$$

o.w.

$$|Y| < x \leq 1$$

$$f_{XY} = f_{Y|X} \cdot f_X$$

$$= \begin{cases} 1 \\ 0 \end{cases}$$

$$0 \leq x \leq 1$$

$$-x \leq Y \leq x$$

o.w.

$$= \begin{cases} 1 \\ 0 \end{cases}$$

$$|Y| < x \leq 1$$

o.w.

$$f_Y = \int_{|Y|}^1 dx$$

$$= 1 - |Y|$$

$$\frac{f_Y(Y)}{f_Y}$$

$$= \begin{cases} 1 - |Y| \\ 0 \end{cases}$$

$$-1 \leq Y \leq 1$$

$$|Y| \leq 1$$

o.w.

$$P(|Y| < x^3)$$

=